

Note that you can refine your testing plan as the project development goes on. Keep the change log as follow:

Change Log

Version	Change Date	By	Description
version number	Date of Change	Name of person who made changes	Description of the changes made
1	Feb 27, 2024	Team8	Create testing plan for sprint 1
2	Apr 9, 2024	Ngoc Kien Mai	Release project

Introduction

Scope

Features:

- Login/Signup functionalities
- Course Look Up: testing selected Term + Department => Courses
- Get Courses/Sections that a User registered
- Roadmap for degree

Non-functional requirements:

- Response time less than 300ms

Roles and Responsibilities

Name	Net ID	GitHub username	Role
Quang Anh Le	7900715	AnhLe-Axel	Developer
Jack Nguyen	7920902	ntt2402vn	Developer
Arshdeep	7917921	arsh331	Developer
Ngoc Kien Mai	7876083	maingockien01	Developer

Test Methodology

Test Completeness

Here you define the criteria that will deem your testing complete. For instance, a few criteria to check Test Completeness would be

- 100% back-end code coverage (mandatory for this project), all the back-end source code should be covered by test cases.

Unit tests

- Look up Courses Test:
 - Test Degree.Controller: Function findAll() should return a list of Degree that are available.
 - Test Term.Controller : Function find(tid, department) should return a list of Courses with matched Deperment in that specific Term. Function findAll() should return a list of all Terms.
 - > Acceptance criteria: Result returned should match with provided array JSON . (based on Database implementation)
- Get Courses registered of a User Test:
 - Test User.Course.Controller: function findActive(uid,tid) should return a JSON array of Courses registered with mached User ID and Term ID.
 - > Acceptance criteria: Result returned should match with provided array JSON of Courses. (based on Database implementation)
- Roadmap for degree
 - DegreeController: Get degree by ID with options to include recommended courses
 - Degree Service: Get degrees with criteria and options for including recommended courses relations
- Authentication Module:
 - Test auth.controller: Function signup() should create a new user.
- Calendar for the user:
 - Test user.course.controller: Function should return all the courses for a user in a term.
 - Acceptance test: Result should match the test courses.

Integration tests

Total number of test cases: 11

1. term.controller.e2e.ts:
 - Test suite: 'UserCourseController'
 - Test case: 'findAll()': This test case is likely testing the functionality of the 'findAll' method in the 'UserCourseController'. The specifics of the test are not provided in the code excerpt.
 - Test case: 'findCurrent()': This test case is likely testing the functionality of the 'findCurrent' method in the 'UserCourseController'. The specifics of the test are not provided in the code excerpt.

2. term.service.e2e.ts:
 - Test suite: 'TermService'
 - o Test case: 'findAll()': This test case is testing the 'findAll' method in the 'TermService'. It expects the method to return an array.
3. userCourse.controller.e2e.ts:
 - Test suite: 'UserController'
 - o Test case: 'searchSection': This test case is testing the 'searchSection' method in the 'UserController'. It checks for unauthorized access, allows registered users to access the function, and expects the function to return an array for registered users.
 - o Test case: 'searchActive': This test case is testing the 'searchActive' method in the 'UserController'. It checks for unauthorized access, allows registered users to access the function, and expects the function to return an array for registered users.
 - o Test case: 'add': This test case is testing the 'add' method in the 'UserController'. It checks for unauthorized access and expects an exception for invalid section id.
 - o Test case: 'remove': This test case is testing the 'remove' method in the 'UserController'. It checks for unauthorized access.
4. userCourse.service.e2e.ts:
 - Test suite: 'UserService'
 - o Test case: 'findAll()': This test case is testing the 'findAll' method in the 'UserService'. It expects the method to return an array and contain a specific user.
 - o Test case: 'find()': This test case is testing the 'find' method in the 'UserService'. It expects the method to find and return a specific user or return null.
 - o Test case: 'findSection()': This test case is testing the 'findSection' method in the 'UserService'. It expects the method to return an empty array or an array type.
 - o Test case: 'findActive()': This test case is testing the 'findActive' method in the 'UserService'. It expects the method to return an empty array or an array type.

Load testing:

The Load Testing was tested with 120 users concurrently using the system. Load Testing includes:

Core features:

Account Management:

1. Sign up request
2. Log in request
3. View profile request
4. Modify profile request

Course Management:

1. View Course look up page request
2. Filter/Search courses based on selected Degree and Term request
3. View Course detail request
4. Add Course request
5. Fetch active registration request (used for later usage of dropping course)
6. Remove/Drop Course request

Calendar:

1. View Calendar page request
2. Fetch active registration request (used for displaying registered courses)

Roadmap and Personal Roadmap:

1. View Roadmap page request
2. View specific degree Roadmap request
3. View Personal Roadmap page request
4. Fetch all courses to display in personal roadmap request
5. Add/Drop courses to Personal Roadmap request

Acceptance tests**Acceptance Tests for Account Management feature****1. Signup Page Navigation Test Steps:**

- Step 1: Open the website.
- Step 2: Click on the "Register" button. Expected Result: The user is directed to the signup page.

2. Signup Form Submission Test Steps:

- Step 1: On the signup page, fill in valid information including username, full name, and password.
- Step 2: Click on the "Signup" button. Expected Result: The user is redirected to the login page.

3. Login Page Navigation Test Steps:

- Step 1: On the login page, ensure the URL is correct. Expected Result: The URL should match the login page URL.

4. Login Form Submission Test Steps:

- Step 1: On the login page, enter the username and password provided during signup.

- Step 2: Click on the "Login" button. Expected Result: The website should authenticate the user's credentials and redirect them to their home page.

5. Home Page Display After Login Test Steps:

- Step 1: After logging in, verify that the user is redirected to their home page. Expected Result: The home page should display personalized content for the logged-in user.

6. Error Handling - Invalid Credentials Test Steps:

- Step 1: On the login page, enter invalid credentials (incorrect username/password).
- Step 2: Click on the "Login" button. Expected Result: The website should display an error message indicating invalid credentials and prompt the user to try again.

7. Error Handling - Missing Fields on Signup/Login Test Steps:

- Step 1: On the signup/login page, leave one or more required fields (username, full name, password) blank.
- Step 2: Click on the "Signup" button. Expected Result: The website should display error messages indicating the missing fields and prompt the user to fill them in.

8. Accessing User Profile Page Test Steps:

- Step 1: After logging in, click on the user icon.
- Step 2: From the drop-down menu, select the "Profile" option.
- Step 3: On the User Profile page, verify that the current username and full name are displayed. Expected Result: The user is navigated to the User Profile page with their username and full name visible.

9. Editing User Information Test Steps:

- Step 1: On the User Profile page, locate the fields for username, password, and full name.
- Step 2: Enter new information into one or more of the fields (e.g., new username and/or password and/or full name).
- Step 3: After entering new information, click on the "Update" button.
- Step 4: Verify that the updated user information (new username and/or password and/or full name) is displayed on the User Profile page. Expected Result: The User Profile page reflects the changes made by displaying the updated user information.

10. Error Handling - Invalid Inputs Test Steps:

- Step 1: Attempt to update the profile with invalid inputs (e.g., an empty username or password that doesn't meet the requirements).
- Step 2: Click on the "Update" button. Expected Result: The website should display appropriate error messages and prevent the update if the inputs are invalid.

Acceptance Tests for Calendar feature

Requirements

1. As a user, I want to view my calendar so that I know my class time.
2. As a user, I want to export my class schedule.

Acceptance Criteria

1. User can see the class timings in the calendar which are in sync with the active courses.
2. User can easily navigate to the export button and can download the timetable as a .ics file.

Tests

1. Check classes on calendar

Test Scenario:

- Navigate to the login page
- Select register option
- Fill up the registration form with the data: Username: "test_user", Full Name: "Test User", Password: "test_user".
- Next, login using these details.
- Navigate to the calendar.
- Select the Add/Drop Courses on the navigation bar.
- Enter "1" as section ID and click Add.
- Move to calendar page.

Expected outcome:

- Initially the calendar is empty and after adding the section, user can see the "Introduction to Probability" course in the calendar.

Actual outcome:

- User sees an empty calendar initially and then after adding the section, "Introduction to Probability" course is visible in the calendar.

Status: **Pass**

Comments: No issues found

2. Export calendar

Test Scenario:

- Navigate to the login page.
- Login using: Username: "test_user" and Password: "test_user".
- Move to the calendar page.
- Click "Export" button on the top right of the calendar.

Expected outcome:

- A file named "schedule.ics" is downloaded, which can be opened in a calendar application.

Actual outcome:

- "schedule.ics" file is downloaded which can be opened in a calendar application.

Status: **Pass**

Comments: The .ics file can be loaded on a calendar application and it correctly reflects the class timings.

Acceptance Tests for Course Management feature

Preconditions:

- User has valid login credentials.
- User has access permissions to view courses.
- At least one term and multiple departments with courses available in the system.
- User has access permissions to add and drop courses functionality.

Test Scenarios:

1. Login to the System:

- Open the application's login page.
- Enter valid login credentials (username and password).
- Click on the "Login" button.
- Verify that the login is successful, and the user is redirected to the Main page.

2. Navigate to the "Look Up Courses" Section:

- Locate and click on the "Courses Look Up" on the navigation bar.

Expected Result: User is redirected to the Course Look Up page. The page should contain 2 Menus of Term and Degree.

3. Select a Term and Degree:

- Choose a specific term and degree from the available options in the List menus. (Degree Menu on the left and Term menu on the right)
- The latest (current) term is the bottom most term displayed in the Menu.
- On selected Term and Degree, the selected item will be colored in RED.
- After selecting terms and degrees, press the “Apply” button below the Menus.

Expected Result: If not both Term and Degree have been selected, click “Apply” button will do nothing and stay at the current page. If both Term and Degree have been selected by user, click “Apply” button below the Menus, it will redirect users to a page containing a list of courses that match selected term and degree.

4. Course Display in List:

- Courses display should contain Course ID, Name of the course, Course Number, Description of the course from left to right.
- Can scroll up and down in the List to find the courses displayed.
- Click on the specific course of user’s choice to navigate to the detail page of that course.

Expected result: User is redirected to the detail page of the clicked on Course, which contain more specific information about the course like Time, Sections.

5. See the Course Information/Detail:

- When clicked on a specific course on the List, the detail page will appear, in which contains information about that course: Course Name, Department, Course Number, Description, Prerequisites (Courses need to be completed before registering the current course), and Sections – this contains Section ID, Professor who will teach the course, Location of the course, Time, and Term of that Section.

Expected Result: The Term displayed in the section should be the Term that was selected by user. The Department shown in the information of the course should match the degree that User selected.

6. Add Courses through Detail page:

- If the selected term is the current term (Fall 2024)(in the Select Term and Display section above), the courses in the display list should be able for user to register.
- The detail page of these courses should contain Sections that have a “Add Section” Button right next to the right of each section display. If the term selected is not the current term, then the detail page is as described in the “See the Course Information/Detail” with no “Add Section” Button next to each sections.
- Click on the “Add Section” button to add the selected section of the specific course chosen to the user calendar.

Expected Result:

- Case 1: User is not currently registered in the selected Section or any other sections that are related to the Course selected and has done all prerequisites of the course

User will be redirected to the add/drop page, which will then display the section/course just registered by the User below the Registered Sections. It will display the Course ID, Section Name, Location , Time from left to right and a wastebasket icon right next to each registered section.

- Case 2: Duplicate Section registration

There will be an error red notification appear on the left bottom side of the screen says that “You have already registered for this section”, and the user will not be redirected to the add/drop page, the add fails, user stays at the current page.

- Case 3: Duplicate Course registration (Registering 2 Sections of the same Course)

There will be an error red notification appear on the left bottom side of the screen says that “You have already registered for this course”, and the user will not be redirected to the add/drop page, the add fails, user stays at the current page.

- Case 4: User has not finished all prerequisites required by the selected course

There will be an error red notification appear on the left bottom side of the screen says that “User has not completed all the prerequisites to register this course ”, and the user will not be redirected to the add/drop page, the add fails, user stays at the current page.

- Case 5: Overlapped in time with registered course

There will be an error red notification appear on the left bottom side of the screen says that “Time overlapped with registered course ”, and the user will not be redirected to the add/drop page, the add fails, user stays at the current page.

7. Add/Drop Courses page:

- From the Main page, or from any page, locate and click on the “ADD/DROP COURSES” Button on the Navigation Bar.

Expected Result: User should be redirected to an add/drop page which contains Add Section text box with the “ADD” button next to it, and right under it is the Registered Sections which contains all registration of the user in the current Term (Fall 2024).

8. Add Courses through Add/Drop Courses page:

- Fill in the Text Box under the Add Section with a specific Section ID.
- After filling the input box of specific section ID, press the “ADD” button next to the right of it. Specific section ID can be found via Detail Page of a specific course as described in the “See the Course Information/Detail” above.

Expected Result:

- Case 1: User is not currently registered in the selected Section or any other sections that are related to the Course selected and has done all prerequisites of the course

The Section ID filled in the input box by the user will disappear. The section/course just registered by the User will immediately display below the Registered Sections (it means that user has successfully registered on that course/section). The registered section/course will display the Course ID, Section Name, Location , Time from left to right and a wastebasket icon right next to each registered section.

- Case 2: Duplicate Section registration

There will be an error notification appear right under the input box saying that “You have already registered for this section”, the add fails, user stays at the current page and there will be no new courses appear under the registered sections.

- Case 3: Duplicate Course registration (Registering 2 Sections of the same Course)

There will be an error notification appear right under the input box saying that “You have already registered for this course”, the add fails, user stays at the current page and there will be no new courses appear under the registered sections.

- Case 4: User has not finished all prerequisites required by the selected course

There will be an error notification appear right under the input box saying that “User has not completed all the prerequisites to register this course”, the add fails, user stays at the current page and there will be no new courses appear under the registered sections.

- Case 5: Overlapped in time with registered course

There will be an error notification appear right under the input box saying that “Time overlapped with registered course”, the add fails, user stays at the current page and there will be no new courses appear under the registered sections.

- Case 6: Invalid section ID

There will be an error notification appear right under the input box saying that “Section ID does not exist”, the add fails, user stays at the current page and there will be no new courses appear under the registered sections.

9. Drop Courses through Add/Drop Courses Page:

- In the add/drop courses page, if the user has registered in some courses, the registered section will not be empty. . It will display each course registered with Course ID, Section Name, Location , Time from left to right and a wastebasket icon right next to each registered section. This wastebasket icon represents for the deletion of a course.
- When click on the wastebasket icon of a chosen course that user wish to delete from their calendar (dropping the course in the system), the pop up text will appear, which ask for confirmation from user, displaying : “ Confirmation Deletion, Are you sure you want to drop this section” with 2 button “Cancel” and “Drop” respectively under the pop-up text.
- Click on Cancel or outside the pop-up text to cancel the behavior of dropping, click on Drop if decide to drop the course.

Expected Result:

- Click “Cancel” or outside the pop-up text: The Pop-up text disappears, user returns back to the Add/drop courses page with no changes to the Registered Sections.
- Click “Drop”: The Pop-up text disappears, user returns back to the Add/Drop Courses page with the selected course deleted in the Registered Sections (e.g. There are 2 courses before dropping, after clicking Drop on the “Course 1”, user returns back the Add/Drop Course page with no “Course 1” under the registered Section since it had been deleted).

10.Error Handling:

- Test for error scenarios such as selecting invalid section IDs.
- Verify that appropriate error messages are displayed when invalid selections are made.
- Ensure that the system gracefully handles errors without crashing or malfunctioning

Acceptance Tests for Roadmap feature

Pre-step: Log in as user

Degree roadmap

- Step 1: Open the website
- Step 2: Click Roadmap button on Navigation bar
- Step 3: Click on Select Degree input
- Step 4: After list of degrees drops down, click on the degree
- Step 5: Expecting a list of recommended courses show up
- Step 6: Hover on courses, selected course will be purple, while prerequisites will have blue

Personal roadmap

- Step 1: Open the website
- Step 2: Click Personal Roadmap button on Navigation bar
- Step 3: Expecting a list of picked courses show up. Empty if users have no courses picked.
- Step 4: Click on Select a course to see the course
- Step 5: Select a course from the drop down
- Step 6: See courses
- Step 7: Click add to Roadmap
- Step 8:

Case 1: Successful. New courses show up in roadmap

Case 2: Prerequisites requirements does not meet, alert error.

Case 3: Courses already existing in roadmap, alert error.

- Step 9: Click on x button on course chip to remove course from roadmap
- Step 10:

Case 1: Course is prerequisite for others in roadmap, alert error.

Case 2: Successful, courses are removed.

- Step 11: Click on button "Save" to save the roadmap

Resource & Environment Needs

Testing Tools

- **Jest** for unit tests in backend & frontend
- **Supertest** for integration tests in backend
- **GitHub Actions** for running tests in CI
- **GitHub Issues** to keep track of bugs, tasks, ...
- **Postman** for testing backend API responses
- **TablePlus/SQL workbench** for troubleshooting database
- **ES-Lint** for linting the code
- **Locust** for load testing

Test Environment

- Windows 10 and above with installed WSL (Ubuntu)
- Docker (integrated with WSL for Windows)
- Docker image:
 - Node 18
 - Global yarn packages required: typescript, @nestjs/cli, concurrently

Terms/Acronyms

Make a mention of any terms or acronyms used in the project

TERM/ACRONYM	DEFINITION
API	Application Program Interface
AUT	Application Under Test